



OPEN Transfer learning-based SOH estimation from partial charging data with polarization-aware modeling

Yongxu Li¹, Yuanyuan Gao², Xianbao Wang², Cui Wang², Zhenyi Wang², Chuxiong Wu² & Jiali Wang²

Accurate state-of-health (SOH) estimation from incomplete charging data remains difficult when batteries are operated under variable charging protocols and when the prior polarization history is unknown. Many existing data-driven methods either rely on handcrafted health indicators or assume relatively fixed sampling conditions, which limits their robustness when charging segments differ in location, duration, and current profile. In this study, we develop a transfer-learning framework in which a long short-term memory (LSTM) network is first pretrained on a voltage-prediction task and is then fine-tuned for SOH estimation. The motivation is that accurate voltage prediction requires the hidden state to encode the latent polarization dynamics that connect recent current-voltage history to future battery response. Using the public Severson fast-charging dataset of 124 commercial lithium iron phosphate/graphite cells, we evaluate the proposed framework on 20%-capacity charging segments sampled at varying starting positions. Under the matched in-house protocol used in this study, transfer learning reduces the SOH estimation mean absolute error from 1.75% to 0.91% and the root mean square error from 2.35% to 1.30% relative to direct LSTM training. We further clarify the comparison protocol, discuss the scope and limitations of cross-paper comparisons, and position the method against recent partial-charging, CNN-based, Transformer-based, and transfer-learning studies. The results support the value of mechanism-informed pretraining for improving data efficiency and robustness, while also showing that broader benchmark reruns under a unified protocol remain an important next step for future work.

Keywords Lithium-ion battery, State of health, Partial charging, Transfer learning, LSTM, Polarization state

Lithium-ion batteries are widely used in electric vehicles, portable electronics, and stationary energy-storage systems because of their high energy density, long cycle life, and favorable power capability. However, their performance gradually degrades during operation because of coupled physicochemical aging mechanisms, including loss of cyclable lithium, growth of the solid electrolyte interphase, structural degradation of active materials, electrolyte decomposition, and, under harsh charging conditions, lithium plating¹. Reliable state-of-health (SOH) estimation is therefore essential for safety management, maintenance scheduling, and lifetime-aware control². In this work, SOH is defined as the ratio between the current chargeable capacity and the nominal capacity.

A substantial body of research has explored SOH estimation using measurable electrical signals such as voltage, current, temperature, and capacity throughput. Existing methods can broadly be divided into model-based approaches, feature-based data-driven approaches, and end-to-end learning approaches. Recent review articles have emphasized that, despite rapid progress, several challenges remain unresolved, including robustness to irregular sampling, sensitivity to preprocessing and feature selection, limited cross-condition generalization, insufficiently fair benchmarking across studies, and the trade-off between predictive accuracy and physical interpretability. These issues are especially important in practical battery management systems, where complete charge records are often unavailable and high-quality labeled degradation data are expensive to obtain^{3,4}.

¹College of Arts and Information Engineering, Dalian Polytechnic University, No. 117, Xuefu Avenue, Zhuanghe, Liaoning, China. ²School of Mechanical and Electrical Engineering, Qingdao City University, No. 79, Tiejishan Road, Qingdao, Shandong, China. ✉email: boklee@126.com; jiali.wang@qdc.edu.cn

Traditionally, battery SOH has often been evaluated from complete charging or discharging curves under relatively controlled laboratory conditions. In practical applications such as electric vehicles and large-scale energy-storage systems, however, charging often starts from arbitrary states of charge and ends before a full charge is completed. This practical constraint has motivated growing interest in SOH estimation from partial charging data. One line of research extracts handcrafted health indicators from selected voltage windows or curve segments, such as incremental-capacity peaks, differential-voltage features, partial-voltage descriptors, integrated-voltage indicators, or other curve-shape statistics. Representative studies include micro-voltage-interval SOH estimation based on ICA peak distribution⁵, flexible SOH estimation from partial charging segments⁶, partial-voltage-curve feature mining with convolutional models⁷, direct SOH estimation from partial charging data⁸, health prognostics based on partial incremental-capacity analysis⁹, and SOH estimation of cycle-aged cells using partial charging information¹⁰. These studies have demonstrated that even incomplete charging segments can contain useful degradation information when appropriate features are extracted.

In parallel, another major line of research has increasingly relied on end-to-end sequence learning to reduce dependence on manual feature design and to learn directly from charging sequences. Convolutional neural networks, recurrent neural networks, gated sequence models, and Transformer-based architectures have all been applied to battery SOH estimation from partial or short charging snippets. Representative examples include deep-learning-based flexible SOH and SOC estimation from partial charging data¹¹, prediction based on randomly selected partial charges¹², multi-cycle charging-information-guided SOH estimation using pretrained sequence models¹³, Transformer-LSTM fusion models¹⁴, generalized transfer-learning frameworks for cross-condition SOH estimation¹⁵, deep transfer learning using short-term charging data¹⁶, and recent fast-charging-oriented deep learning frameworks based on partial charge information¹⁷. These developments reflect a broader shift in battery prognostics toward learned temporal representations rather than purely handcrafted indicators.

Despite this progress, an important difficulty remains insufficiently emphasized in many partial-charging studies: the measured charging voltage is influenced not only by the present current and the long-term degradation state, but also by the latent internal dynamic state of the battery. Even when two charging segments follow similar external current protocols, their voltage trajectories may differ because the internal polarization state depends strongly on recent operating history and relaxes over time. The importance of polarization and relaxation behavior has long been recognized in electrochemical battery modeling^{18,19}, and recent studies have shown that relaxation signals can help reveal degradation-related phenomena such as lithium plating²⁰. In addition, variable fast-charging strategies further increase the complexity of voltage response by altering the interplay between charge-transfer, transport limitation, and degradation evolution^{21,22}. Therefore, partial charging data contain not only degradation information but also transient history-dependent electrochemical information that is not directly observable in a battery management system.

This observation motivates the central idea of the present study. Rather than training a model to infer SOH directly from a partial charging segment alone, we first pretrain a recurrent network on a voltage-prediction task and then transfer the learned representation to SOH estimation. The rationale is that voltage prediction provides dense supervision at every time step, whereas SOH labels are sparse at the cycle level. If the recurrent layers have already learned a representation that captures the hidden dynamic factors governing short-term voltage evolution, fine-tuning for SOH estimation may become more data-efficient and more robust than direct training from scratch. In this sense, the proposed framework is not merely a generic use of transfer learning, but a mechanism-informed strategy in which the source task is selected to encourage the network to internalize physically meaningful temporal structure.

The use of a long short-term memory (LSTM) network is particularly suitable in this context because its hidden state and cell state provide a compact representation of recent temporal history^{23,25}. Although the LSTM state is not interpreted here as a one-to-one electrochemical state variable, its gated memory-update mechanism is qualitatively consistent with the way polarization accumulates and relaxes under changing inputs²⁴. This design therefore introduces a physically motivated inductive bias into a data-driven architecture while avoiding the strong assumptions required by explicit electrochemical parameter identification. More broadly, the growing success of temporal deep-learning models in energy and sequence domains suggests that recurrent and attention-based architectures can serve as effective tools for extracting degradation-related information from incomplete measurements²⁶.

To evaluate this idea, we use the public dataset released by Severson et al.²⁷, which contains cycling data from 124 commercial lithium iron phosphate/graphite cells aged under diverse fast-charging protocols. This dataset is widely used because it provides substantial diversity in charging conditions and cycle life, making it a meaningful benchmark for testing whether models trained on partial charging segments can generalize across heterogeneous charging histories. At the same time, published results on this dataset are often difficult to compare directly because train/test partitions, preprocessing pipelines, sampling strategies, targets, and evaluation metrics differ across studies. We therefore position our contribution primarily through a controlled in-house comparison under a matched protocol, while using prior literature as broader context rather than as a strict apples-to-apples benchmark.

In summary, this study proposes a transfer-learning framework for lithium-ion battery SOH estimation from partial charging data, in which an LSTM is first pretrained for voltage prediction and then fine-tuned for SOH regression. The scientific hypothesis is that accurate voltage prediction requires the network to encode latent polarization-related history information that is also beneficial for downstream SOH estimation. Compared with traditional feature-engineering methods, the proposed framework reduces reliance on manually designed descriptors; compared with direct end-to-end SOH regressors, it exploits a dense auxiliary supervision signal before fine-tuning; and compared with generic transfer-learning approaches, it selects a source task with a clearer physical motivation. The remainder of this paper presents the methodological framework, the dataset and experimental protocol, and the empirical evaluation under variable partial-charging conditions.

Constant d	0.0022
LSTM Hidden states h_i dimension	64
LSTM Cell states C_i dimension	64
Embedding W_1	128*3
Embedding W_2 size	32*128
Dimensionality of FC layers for voltage prediction	128
Dimensionality of FC layers for SOH estimation	128
Learning rate	0.001
Max gradient 2-norm of LSTM model for voltage prediction	0.0005
Max gradient 2-norm of FC network for voltage prediction	0.0005
Batch size	20,000

Table 1. Model hyperparameters.

Methodology

Battery mechanism and the role of polarization

The measured charging voltage of a lithium-ion battery is jointly affected by current, available capacity, SOH, and the instantaneous polarization state. In this paper, polarization state denotes the transient deviation from electrochemical equilibrium associated with charge-transfer, ion-transport, and concentration effects. It depends strongly on the recent operating history and gradually relaxes when external excitation is removed. Because this hidden state is not directly measured, two charging segments with the same nominal current protocol may still exhibit different voltage trajectories if their prior histories differ.

This hidden-state perspective is important for partial-charging SOH estimation. In practical battery systems, current and voltage are routinely measured and capacity throughput can be approximately tracked by ampere-hour integration, whereas polarization cannot be observed directly. A model that ignores the latent dynamic state must explain voltage variation only through explicit inputs, which can degrade generalization when charging segments start from different internal conditions. Our approach therefore seeks to encode polarization information implicitly in a recurrent state rather than through hand-engineered features.

Mechanism-informed learning with LSTM

LSTM networks are well suited to this setting because their cell state acts as a filtered memory of recent inputs. The forget and input gates control how quickly previous information decays and how new information is integrated. This behavior resembles the relaxation and update of polarization: recent excitation has a stronger effect than distant excitation, and states that are far from equilibrium tend to return more rapidly toward a stable region. We do not claim that the LSTM cell state is a direct electrochemical state variable; instead, we use it as a compact latent representation whose temporal behavior is consistent with the qualitative behavior expected from polarization dynamics.

Based on this interpretation, the recurrent state is intended to carry both slowly varying SOH-related information and rapidly evolving polarization-related information. The hidden state then combines this internal representation with the present input to predict the next-step voltage. This design does not replace physics-based electrochemical modeling, but it embeds a physically motivated inductive bias into a data-driven architecture.

Transfer-learning framework

Direct SOH supervision is sparse because SOH is approximately constant within a cycle, whereas voltage-prediction supervision is dense at every time step. We therefore pretrain the LSTM on a voltage-prediction task using observed current, voltage, and capacity-step sequences. After convergence of the voltage model, the learned recurrent parameters are transferred to the SOH estimation model, and only the output head is replaced and fine-tuned for SOH regression.

The key idea is that good voltage prediction requires the model to encode the hidden dynamic factors that shape voltage evolution. If the recurrent layers have already learned a representation that is informative about these hidden factors, fine-tuning for SOH becomes easier and less data-hungry than training from scratch. This mechanism-informed use of transfer learning is the central methodological claim of the present work.

Experiments and results

Data description and split protocol

We evaluated the method on the public dataset released by Severson et al. [27], which contains cycling data from 124 commercial lithium iron phosphate/graphite cells with a nominal capacity of 1.1 Ah. The cells were aged under multiple fast-charging protocols until end of life, defined as 80% of nominal capacity. The dataset is attractive for this study because it contains substantial protocol diversity rather than a single fixed charging condition, making it suitable for testing whether a model trained on partial charging data can generalize across heterogeneous charging histories.

Charging generally consisted of two constant-current stages before the 3.6 V cutoff, followed by a constant-current/constant-voltage phase to 80% capacity, whereas discharge was performed at 4 C constant current to 2.0 V. The charging protocols varied in the first-stage current C1, second-stage current C2, and switching point

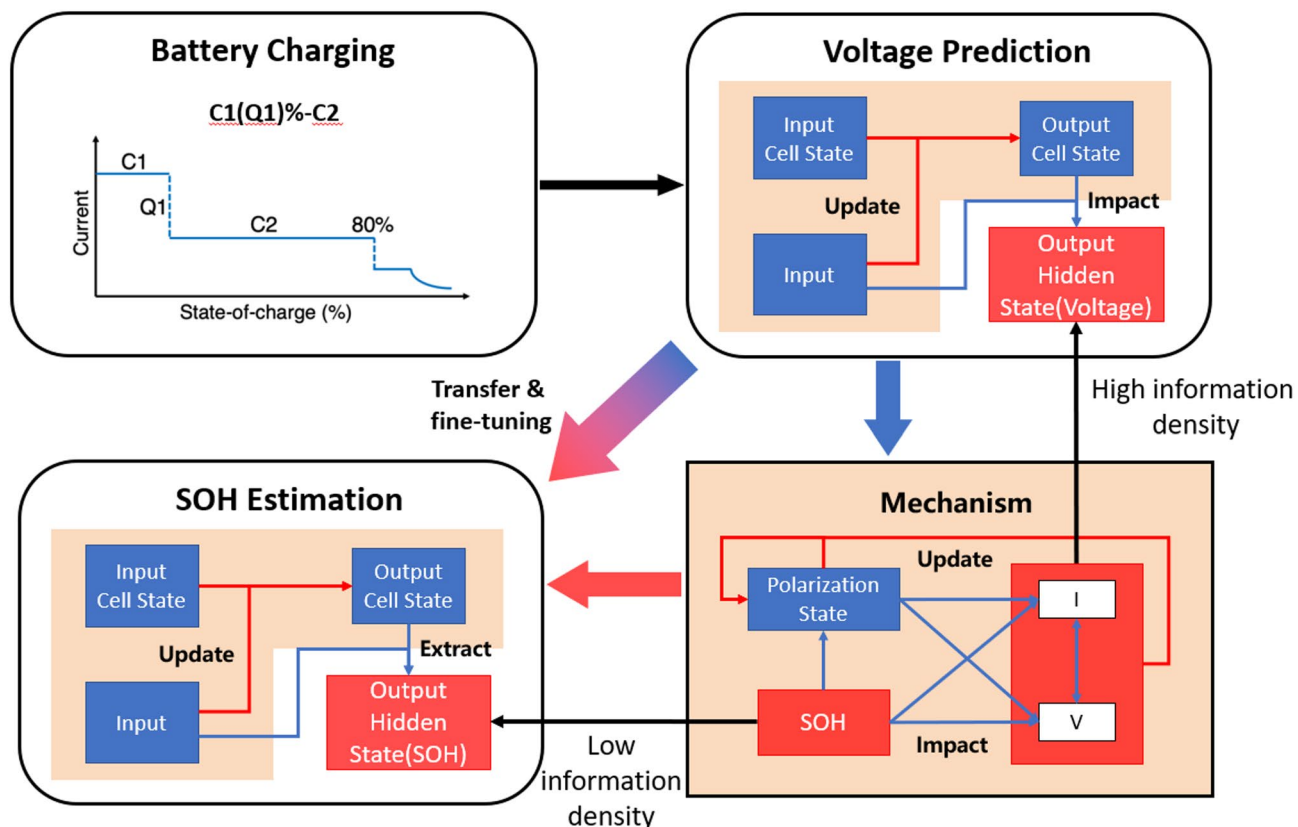


Fig. 1. The transfer learning model. The figure illustrates a transfer learning framework for SOH estimation. During the segmented constant-current charging process ($C1(Q1)\% - C2$), the current-voltage response provides high-density information for training. In the initial stage, the hidden state of the LSTM is responsible for predicting the next-step voltage evolution, while the cell state preserves and updates long-term polarization state and information. The knowledge captured in the hidden state during the voltage prediction task is transferred and fine-tuned for SOH estimation, where the decoupled information between the hidden state and the cell state is leveraged to extract SOH features. The polarization state, SOH, and the current-voltage response mechanism are implicitly coupled within the model, ensuring that the transferred representation aligns with the real physical evolution of the battery. The overall framework is designed for offline training, while the online testing stage utilizes the transferred and fine-tuned model for efficient SOH estimation.

Q1, thereby creating substantial variation in the voltage response observed from short charging segments. All experiments were conducted at 30 °C in temperature-controlled chambers, and the observed cycle life ranged from approximately 150 to 2300 cycles.

Following prior preprocessing practice for this dataset, one extreme outlier with abnormally short life was excluded. The remaining cells were divided into 99 training cells and 24 test cells. The test set included cells cycled under both shared protocols and unique protocols, so the evaluation reflects not only interpolation within known charging patterns but also generalization to less frequently observed protocol settings. Throughout all in-house comparisons in this manuscript, the same train/test split and the same preprocessing pipeline were used.

Experiment design, preprocessing, and fairness protocol.

For each cycle, we generated partial-charging samples with an observed window of 20% nominal capacity. The starting point of the observed window was moved from 0% to 50% nominal capacity in 1% increments, yielding up to 51 samples per cycle after excluding abnormal records. Each 1% capacity interval was interpolated into five equally spaced steps, and the subsequent prediction window for voltage pretraining was set to 7% nominal capacity. Because the charging protocols in the dataset include two constant-current stages followed in many cases by a constant-voltage stage, the resulting samples naturally cover four practical cases considered in this study: one CC stage only, one CC stage followed by CV, two CC stages, and two CC stages followed by CV.

Table 1 summarizes the hyperparameters of the proposed model. The network was first pretrained on the voltage-prediction task for 50 epochs and then fine-tuned on SOH estimation for another 50 epochs. To ensure that the main comparison isolates the effect of transfer learning rather than data handling or optimization choices, the direct-training LSTM baseline used the same train/test split, the same interpolated inputs, the same sample-construction rule, the same optimizer family, the same batch size, and the same total SOH-related training budget, but without voltage-prediction pretraining. The CNN baseline was also trained on the same split and with the same input representation, while the Random Forest baseline used the same per-sample inputs after flattening.

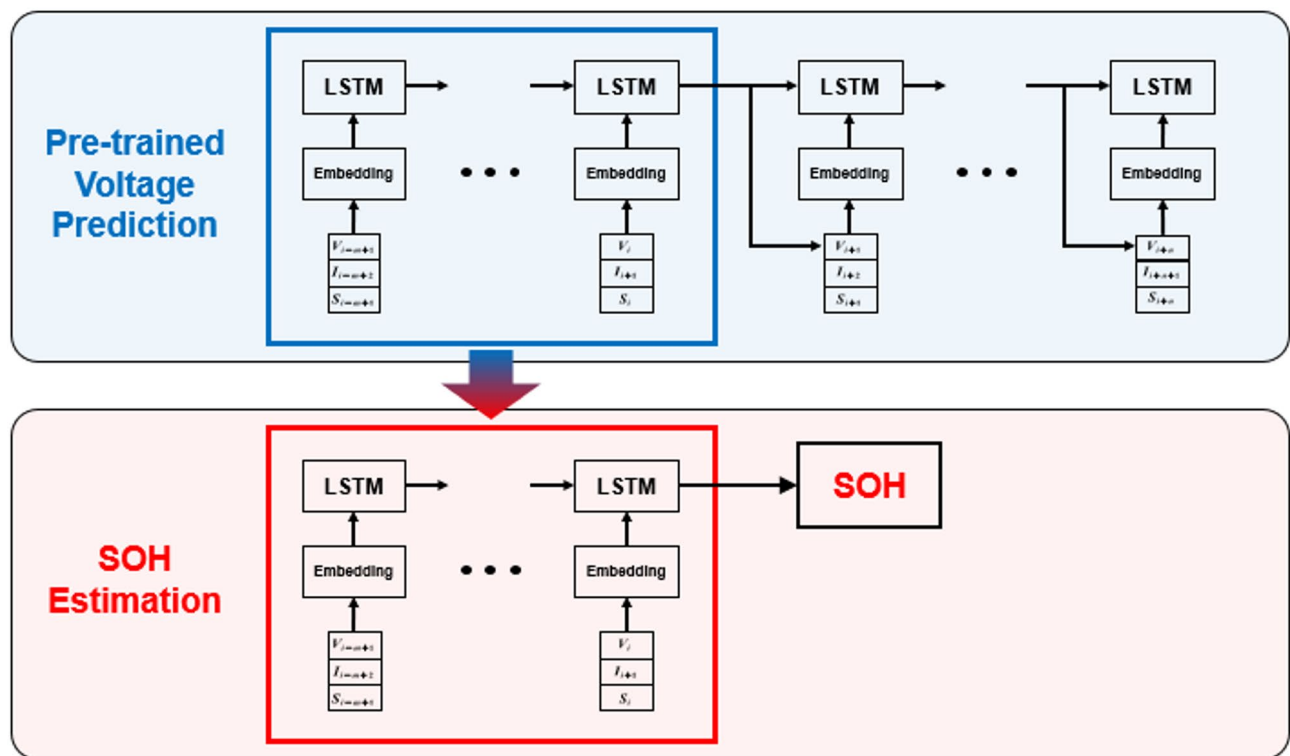


Fig. 2. The transfer learning model. The hidden state is responsible for predicting currents and voltages for the next step, and the cell state primarily preserves and updates the information of polarization state, storage capacity and SOH. The hidden state predicts based on the internal state information from cell state, and the cell state updates based on the new input. The evolution mechanism should be identified and the recognized states in LSTM should regress towards the actual battery internal state as the step proceeds. The battery internal state could be tracked after a sufficiently long charging process. MTL is applied to share the learned information from different tasks. SOH estimation is evaluated by decoupling the internal state information in hidden state and cell state of LSTM. The whole framework is for offline training, and the part in green border is for online testing. The initial stored capacity estimation task is excluded when storage capacity information is provided.

Results and discussion

On the voltage-prediction pretraining task, the model achieved an MAE of 4.3 mV and an RMSE of 10.2 mV, indicating that the recurrent representation learned to track the short-term voltage evolution with good accuracy across cycles and charging protocols. After transfer to SOH estimation, the proposed method achieved an MAE of 0.91% and an RMSE of 1.30%. Under the matched in-house protocol described above, the directly trained LSTM baseline yielded an MAE of 1.75% and an RMSE of 2.35%. These results support the central claim of this work: voltage-prediction pretraining provides a more informative initialization for the recurrent state representation and improves SOH estimation relative to the corresponding LSTM trained directly on SOH labels Fig. 1 and 2.

Figures 3 and 4 provide a qualitative view of where the main errors arise. Larger voltage-prediction errors concentrate near current-switching points, where interpolation is more difficult and the local dynamics change abruptly. However, these local disturbances decay quickly as the sequence progresses, which is consistent with the intended role of the recurrent state in absorbing recent polarization-history information. On the SOH task, most predictions remain tightly clustered around the reference values across cells and charging conditions, while larger deviations tend to appear in more aged cycles or near protocol transitions. In terms of computational cost, training on an NVIDIA Quadro RTX 6000 with a Xeon W-2245 required roughly 3 min per epoch for voltage prediction and 2.5 min per epoch for SOH fine-tuning, which we report to document practical training cost rather than to claim a hardware-independent speed advantage.

We further compared the proposed approach with two commonly used supervised baselines. The 1D CNN achieved an MAE of 0.99% and an RMSE of 1.41%, while the Random Forest achieved an MAE of 1.80% and an RMSE of 2.50%. These additional baselines are helpful for context, although we note that different model families may require different hyperparameter search spaces and training regimes for a fully exhaustive benchmark. Accordingly, we interpret these results as evidence that the proposed approach is competitive and performs best among the in-house settings evaluated here, rather than as proof of universal superiority over all possible alternatives.

Published studies on the Severson dataset differ in at least four aspects that materially affect reported numbers: the prediction target (capacity or SOH), whether results are reported in normalized or percentage units, the

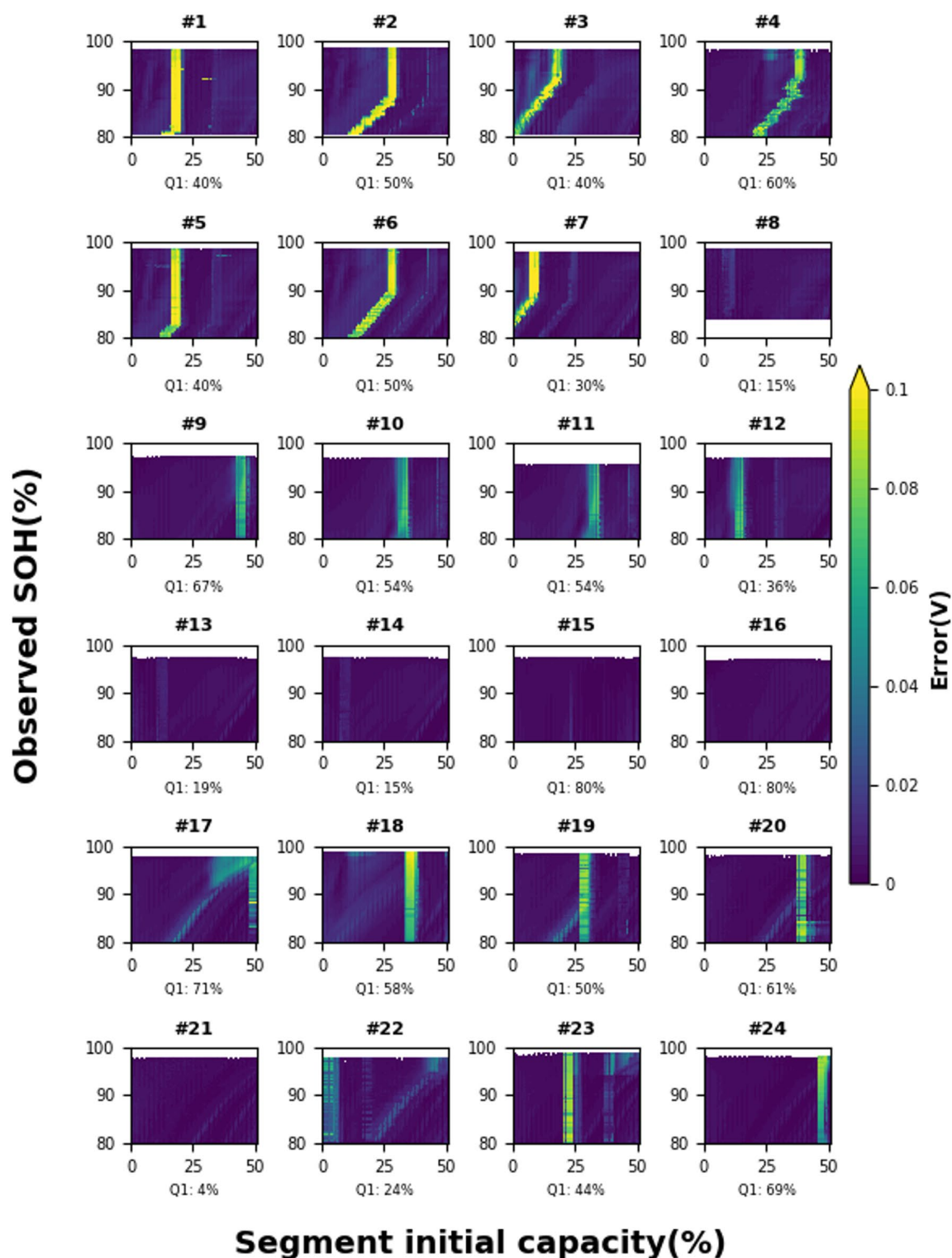


Fig. 3. Voltage prediction error. Voltage-prediction pretraining errors across cycles and charging protocols. The figure highlights that the largest local errors appear near current-switching regions, while most of the sequence remains accurately predicted.

train/test partition, the use of same-battery versus cross-battery prediction, and the input window definition. Therefore, cross-paper numerical comparisons should be interpreted as contextual rather than strictly apples-to-apples unless the codes are rerun under a unified protocol.

Within that limitation, recent published studies provide useful reference points as listed in Table 2. A recent Transformer-based self-supervised SOH estimation study on the 124-cell dataset⁹ reported that, for one

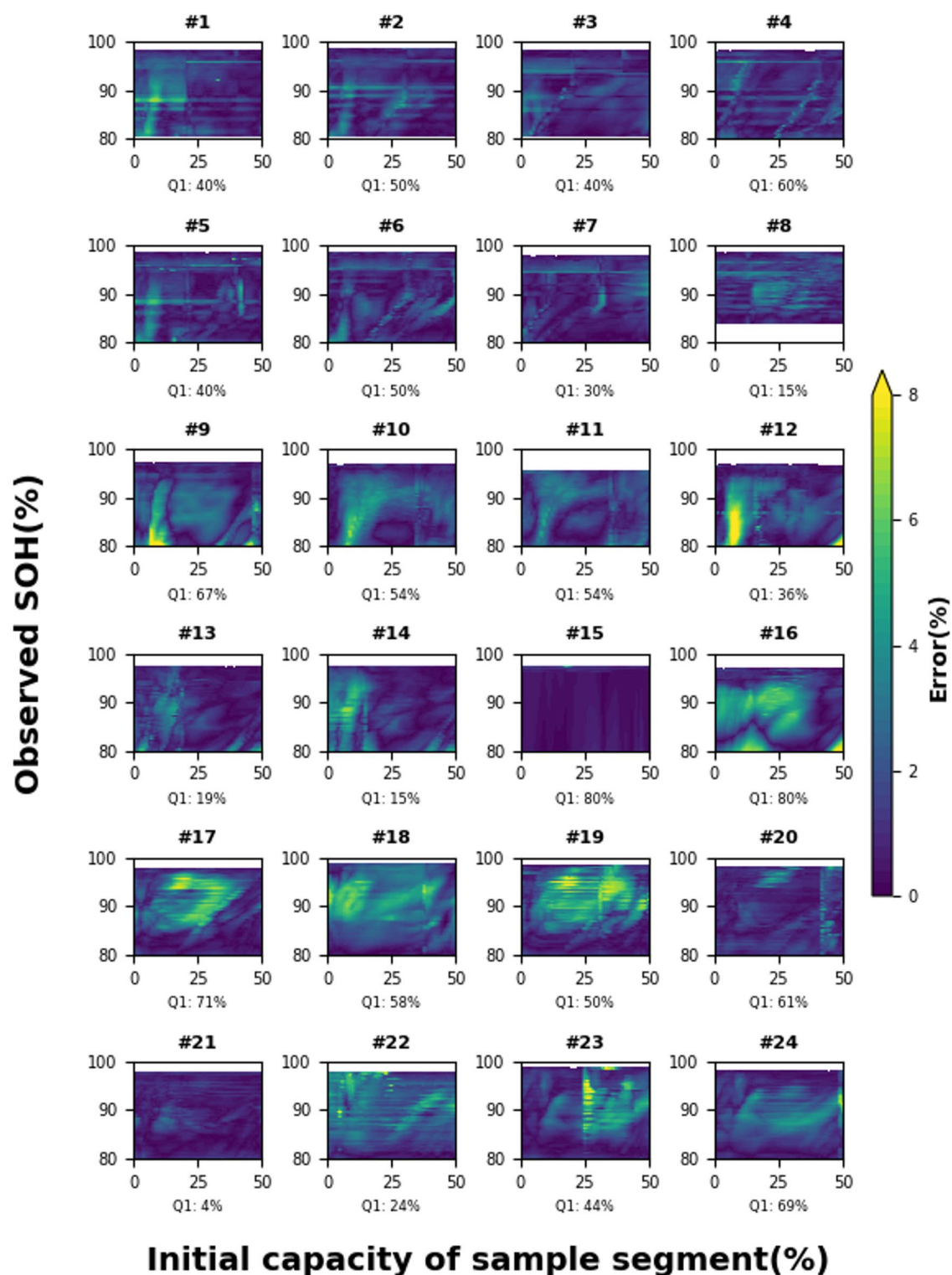


Fig. 4. SOH estimation error. SOH estimation errors across cells and charging protocols. The figure shows that most predictions remain close to the reference SOH values, with larger deviations concentrated in more difficult aging stages or protocol-transition regions.

reported raw-input Transformer setting, MAE and RMSE were approximately 0.98% and 1.07%, respectively, under its own protocol. A recent Transformer-LSTM fusion model evaluated on the same dataset family¹⁹ reported strong performance on selected same-battery and cross-battery settings, but with different train/test definitions and feature engineering. Short charging snippets have also been shown to support flexible SOH and SOC estimation⁶, although that study focused on different datasets and protocol settings. These studies

Study	Dataset relation	Input/protocol	Reported metric(s)	Comparability note
This work	Severson 124-cell dataset	20% capacity partial-charge segments; cross-cell split	MAE 0.91%, RMSE 1.30%	Matched only to the in-house direct-LSTM baseline
Wang et al. 2024	Same 124-cell dataset	Self-supervised Transformer framework	Approx. MAE 0.98%, RMSE 1.07% in one reported raw-input Transformer setting	Different preprocessing and training protocol
Cai et al. 2025	Same dataset family	Feature-engineered Transformer-LSTM fusion	Primary/secondary test-set MAE reported in normalized SOH units	Different feature set, split, and evaluation definition
Tian et al. 2022	Different datasets	Short charging snippets with CNN	RMSE below 12.68 mAh for capacity-related outputs	Different chemistry/protocol and output definition

Table 2. Cross-paper context for recent studies relevant to this manuscript. These results are informative but not strictly comparable unless all methods are rerun under the same protocol.

confirm that the proposed method operates in a highly competitive area and that the key scientific question is not whether learning from partial charge data is feasible, but how to do so robustly under variable conditions and with transparent benchmarking.

Conclusion

This study proposed a transfer learning framework for lithium-ion battery SOH estimation from partial charging data, in which a voltage-prediction pretraining task is used to initialize an LSTM-based SOH estimator. The key idea is not merely to reuse weights from a related task, but to exploit the fact that accurate voltage prediction requires the recurrent representation to encode history-dependent internal information closely related to polarization evolution. Fine-tuning this representation for SOH estimation provides a more informative starting point than direct training on sparse SOH supervision alone.

Experiments on the public Severson dataset 16 show that, under the matched in-house protocol used in this study, the proposed method improves substantially over the direct-training LSTM baseline and also outperforms the CNN and Random Forest baselines that were implemented under the same data split and input construction. These findings support the practical value of mechanism-informed pretraining for SOH estimation from short charging segments under heterogeneous charging strategies. At the same time, the revised manuscript now states this conclusion more carefully: the present experiments demonstrate a clear advantage over the matched in-house baselines considered here, but a broader claim against all prior methods would require additional reruns under a unified protocol.

The study also has important limitations. All experiments were conducted on a single public dataset of LFP/graphite cells, temperature effects and other real-world confounders were not explicitly modeled, and the evaluation relied on a fixed train/test split rather than repeated cross-validation. In addition, although the in-house baseline comparison is now described more rigorously, a more exhaustive benchmark would still benefit from repeated runs, variance estimates, and additional matched baselines such as transformer-based or hybrid architectures. Future work will therefore focus on broader benchmarking, temperature-aware modeling, and extending the framework to other chemistries and operating conditions.

Tables.

Data availability

All data generated or analysed during this study are included in this published article and its supplementary information files.

Received: 29 October 2025; Accepted: 10 April 2026
Published online: 23 July 2026

References

1. Li, Y., Guo, J., Pedersen, K., Gurevich, L. & Stroe, D. I. Investigation of multi-step fast charging protocol and aging mechanism for commercial NMC/graphite lithium-ion batteries. *J. Energy Chem.* **80**, 237–246 (2023).
2. Yang, B. et al. Critical summary and perspectives on state-of-health of lithium-ion battery. *Renew. Sustain. Energy Rev.* **190**, 114077 (2024).
3. Vermeer, W., Mouli, G. R. C. & Bauer, P. A comprehensive review on the characteristics and modeling of lithium-ion battery aging. *IEEE Trans. Transp. Electrification.* **8**, 2205–2232 (2021).
4. Che, Y., Hu, X., Lin, X., Guo, J. & Teodorescu, R. Health prognostics for lithium-ion batteries: mechanisms, methods, and prospects. *Energy Environ. Sci.* **16**, 338–371 (2023).
5. Wang, T. et al. An SOH estimation method within microvoltage interval based on ICA peak distribution for lithium-ion batteries. *IEEE Trans. Transp. Electrification.* **11**, 9224–9233 (2025).
6. Zhang, C. et al. Flexible method for estimating the state of health of lithium-ion batteries using partial charging segments. *Energy* **295**, 131009 (2024).
7. Lu, Z., Fei, Z., Wang, B. & Yang, F. A feature fusion-based convolutional neural network for battery state-of-health estimation with mining of partial voltage curve. *Energy* **288**, 129690 (2024).
8. Gao, R., Zhang, Y. & Lyu, Z. A SOH estimation method of lithium-ion batteries based on partial charging data. *J. Energy Storage.* **103**, 114309 (2024).
9. Meng, H., Geng, M. & Han, T. Long short-term memory network with Bayesian optimization for health prognostics of lithium-ion batteries based on partial incremental capacity analysis. *Reliab. Eng. Syst. Saf.* **236**, 109288 (2023).
10. Jenu, S., Hentunen, A., Haavisto, J. & Pihlatie, M. State of health estimation of cycle aged large format lithium-ion cells based on partial charging. *J. Energy Storage.* **46**, 103855 (2022).

11. Tian, J., Xiong, R., Shen, W., Lu, J. & Sun, F. Flexible battery state of health and state of charge estimation using partial charging data and deep learning. *Energy Storage Mater.* **51**, 372–381 (2022).
12. Vilsen, S. B. & Stroe, D. I. On the use of randomly selected partial charges to predict battery state-of-health. *Batteries* **10**, 193 (2024).
13. Zhang, Z., Zhu, Y., Zhang, Q., Cui, N. & Shang, Y. Multi-cycle charging information guided state of health estimation for lithium-ion batteries based on pre-trained large language model. *Energy* **313**, 133993 (2024).
14. Cai, X. & Liu, T. State of health prediction for lithium-ion batteries using transformer-LSTM fusion model. *Appl. Sci.* **15**, 3747 (2025).
15. Sahoo, S. et al. Transfer learning based generalized framework for state of health estimation of Li-ion cells. *Sci. Rep.* **12**, 13173 (2022).
16. Chen, K. et al. Battery state of health estimation using deep transfer learning on short-term charging data. *Measurement* (2025).
17. Zhao, J. et al. Battery state of health estimation under fast charging via deep transfer learning. *Isience* **28** (5), 112235 (2025).
18. Doyle, M. & Newman, J. The use of mathematical modeling in the design of lithium/polymer battery systems. *Electrochim. Acta.* **40**, 2191–2196 (1995).
19. Jokar, A., Rajabloo, B., Désilets, M. & Lacroix, M. Review of simplified pseudo-two-dimensional models of lithium-ion batteries. *J. Power Sources.* **327**, 44–55 (2016).
20. Chen, X., Li, L., Liu, M., Huang, T. & Yu, A. Detection of lithium plating in lithium-ion batteries by distribution of relaxation times. *J. Power Sources.* **496**, 229867 (2021).
21. Nizam, M., Maghfiroh, H., Irfani, B. & Inayati, I. Ma'arif, A. Designing and prototyping of lithium-ion charging system using multi-step constant current method. *World Electr. Veh. J.* **13**, 178 (2022).
22. Tian, J., Xiong, R., Shen, W., Lu, J. & Yang, X. G. Deep neural network battery charging curve prediction using 30 points collected in 10 min. *Joule* **5**, 1521–1534 (2021).
23. Graves, A. Long short-term memory. In *Supervised Sequence Labelling with Recurrent Neural Networks. Studies in Computational Intelligence* **385**, 37–45 (Springer, 2012).
24. Rehmer, A. & Kroll, A. On the vanishing and exploding gradient problem in gated recurrent units. *IFAC-PapersOnLine* **53**, 1243–1248 (2020).
25. Kanai, S., Fujiwara, Y. & Iwamura, S. Preventing gradient explosions in gated recurrent units. *Adv. Neural Inf. Process. Syst.* **30**, 435–444 (2017).
26. Vaswani, A. et al. Attention is all you need. *Adv. Neural Inf. Process. Syst.* **30**, 5998–6008 (2017).
27. Severson, K. A. et al. Data-driven prediction of battery cycle life before capacity degradation. *Nat. Energy.* **4**, 383–391 (2019).

Acknowledgements

This work was supported by Qingdao Municipal Bureau of science and technology under Grant No. 25-1-1-gjgg-3-gx. We thank for their valuable support.

Author contributions

Yongxu Li conceived the methodology, designed the experiments and wrote the manuscript. Yuanyuan Gao and Xianbao Wang conducted the experiments. Cui Wang, Zhenyi Wang, and Chuxiong Wu contributed to data analysis and interpretation. Jiali Wang supervised the project and provided critical revisions to the manuscript. All authors reviewed and approved the final manuscript.

Funding

The project is supported in part by the Qingdao Municipal Bureau of Science and Technology under Grant 25-1-1-gjgg-3-gx.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-48906-4>.

Correspondence and requests for materials should be addressed to Y.L. or J.W.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026